

Mechatronics and Industrial Internet of Things

IIoT-Based Water Level Detection and Flood Alert System

ASSINGMENT – 10

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PROJECT OVERVIEW

The Water Level Detection and Flood Alert System is an automated monitoring system designed to continuously measure water levels in rivers, tanks, dams, reservoirs, and other water-storage areas. The system uses an ultrasonic sensor to detect the distance between the sensor and the water surface without direct contact with water.

The measured data is processed by a microcontroller, which determines the current water level and compares it with predefined safety limits. The system identifies different conditions such as normal, warning, and critical levels. When the water level rises above the warning limit, the system provides an appropriate indication. If the water reaches the critical or overflow level, an LED and buzzer are activated to provide an immediate flood alert.

The system helps reduce dependence on manual water-level observation and provides faster detection of dangerous water-level changes. It can be used as a simple and low-cost early-warning solution for flood-prone locations.

The proposed system can also be enhanced with Wi-Fi and IoT technology to transmit water-level data to a cloud platform, allowing users to monitor the water level remotely and receive alerts during critical conditions.

PROBLEM STATEMENT

The sudden rise of water levels in rivers, dams, tanks, reservoirs, and drainage systems can cause overflow and severe flooding. Traditional monitoring methods often depend on manual observation, which may not provide continuous information or immediate warnings during rapidly changing water conditions. Delayed detection can result in loss of life, damage to property and infrastructure, and difficulties in taking timely preventive actions. Therefore, an automated system is required to continuously monitor water levels and provide early warnings before critical overflow conditions occur.

- Continuous monitoring of water levels is required in flood-prone areas.
- Manual observation can be time-consuming and may not provide real-time information.
- Rapidly rising water levels can be difficult to detect at an early stage.
- Delayed warnings can increase the risk of damage and loss.
- The system should detect normal, warning, and critical water-level conditions.
- Immediate visual and audible alerts are required during critical conditions.
- A low-cost and reliable solution is needed for practical implementation.
- The system should reduce dependence on continuous human observation.
- The proposed system uses an ultrasonic sensor for non-contact water-level measurement.
- The system can be extended with IoT technology for remote monitoring and mobile alerts.

PROBLEM SOLUTION

The proposed Water Level Detection and Flood Alert System provides an automated and reliable method for continuously monitoring water levels and detecting possible overflow conditions. The system uses an ultrasonic sensor positioned above the water surface to measure the distance between the sensor and the water. Since the measurement is non-contact, the sensor can monitor the water level without being submerged or exposed directly to the water.

The sensor readings are sent to a microcontroller, which processes the data and calculates the corresponding water level. The measured value is continuously compared with predefined safety limits. Based on the detected level, the system identifies whether the condition is normal, warning, or critical.

This allows the system to respond automatically without requiring continuous human observation. During normal conditions, the system continues regular monitoring. When the water level reaches the warning range, a visual indication is provided to alert the user that the water level is increasing. If the water level reaches the critical or overflow threshold, the red LED and buzzer are activated to provide an immediate visual and audible flood warning. This gives nearby users additional time to take preventive or emergency actions.

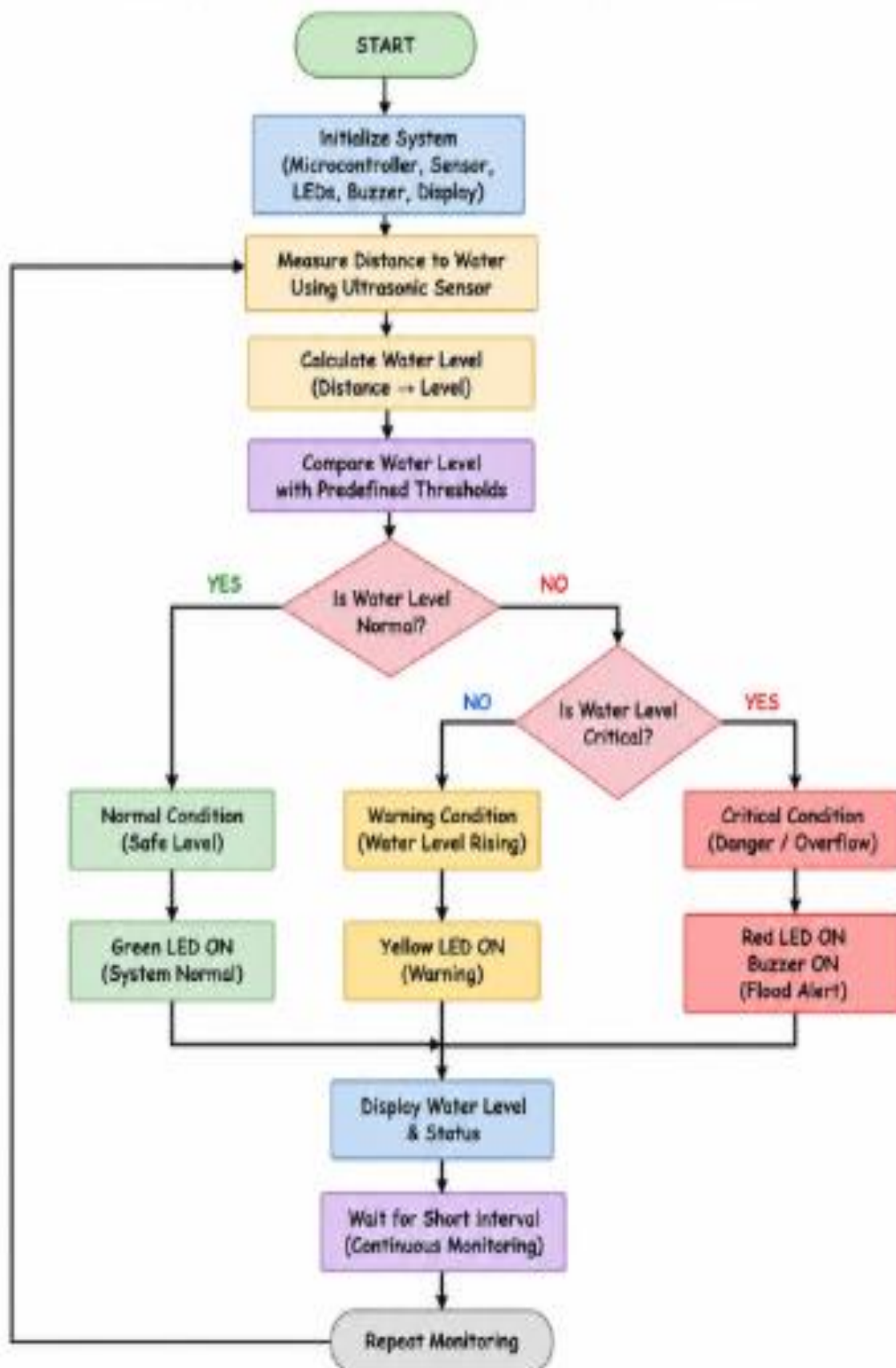
- Provides continuous real-time water-level monitoring.
- Uses an ultrasonic sensor for accurate non-contact measurement.
- Automatically detects rising and falling water levels.
- Reduces the need for continuous manual observation.
- Identifies normal, warning, and critical water-level conditions.
- Provides early warning before the water reaches the overflow condition.
- Activates visual indicators according to the detected water level.
- Activates an audible buzzer during critical flood conditions.
- Helps reduce the risk of damage to people, property, and infrastructure.
- Provides a simple and low-cost solution for flood-prone areas.
- Can be installed in rivers, dams, tanks, reservoirs, canals, and drainage systems.
- Can operate continuously with repeated sensor measurements.
- Can be integrated with a display to show real-time water-level information.
- Can be upgraded with Wi-Fi and IoT technology for remote monitoring.
- Cloud storage can be added to maintain historical water-level records.
- Mobile notifications can be incorporated for immediate remote alerts.
- The system can support faster decision-making during emergency situations.
- The prototype can be expanded for smart-city and disaster-management applications.

SYSTEM ARCHITECTURE

The system starts by initializing the microcontroller, ultrasonic sensor, LED indicators, buzzer, and display. The ultrasonic sensor continuously measures the distance between the sensor and the water surface. The microcontroller processes this measurement and calculates the corresponding water level. The calculated water level is then compared with predefined normal, warning, and critical thresholds. If the water level is normal, the green LED indicates a safe condition. If the water level enters the warning range, the warning indicator is activated to show that the water level is rising. When the water reaches the critical or overflow level, the red LED and buzzer are activated to provide an immediate flood alert. The water level and current status are displayed, after which the system waits for a short interval and repeats the monitoring process continuously.

The system also continuously monitors changes in the water level and updates the displayed information in real time. If the water level decreases, the system automatically returns to the corresponding safe or warning condition and deactivates the critical alert when appropriate. This automatic monitoring and response allow the system to operate without continuous human intervention. The collected water-level information can also be used for analysis and early decision-making, helping users take preventive measures before a possible overflow or flood occurs.

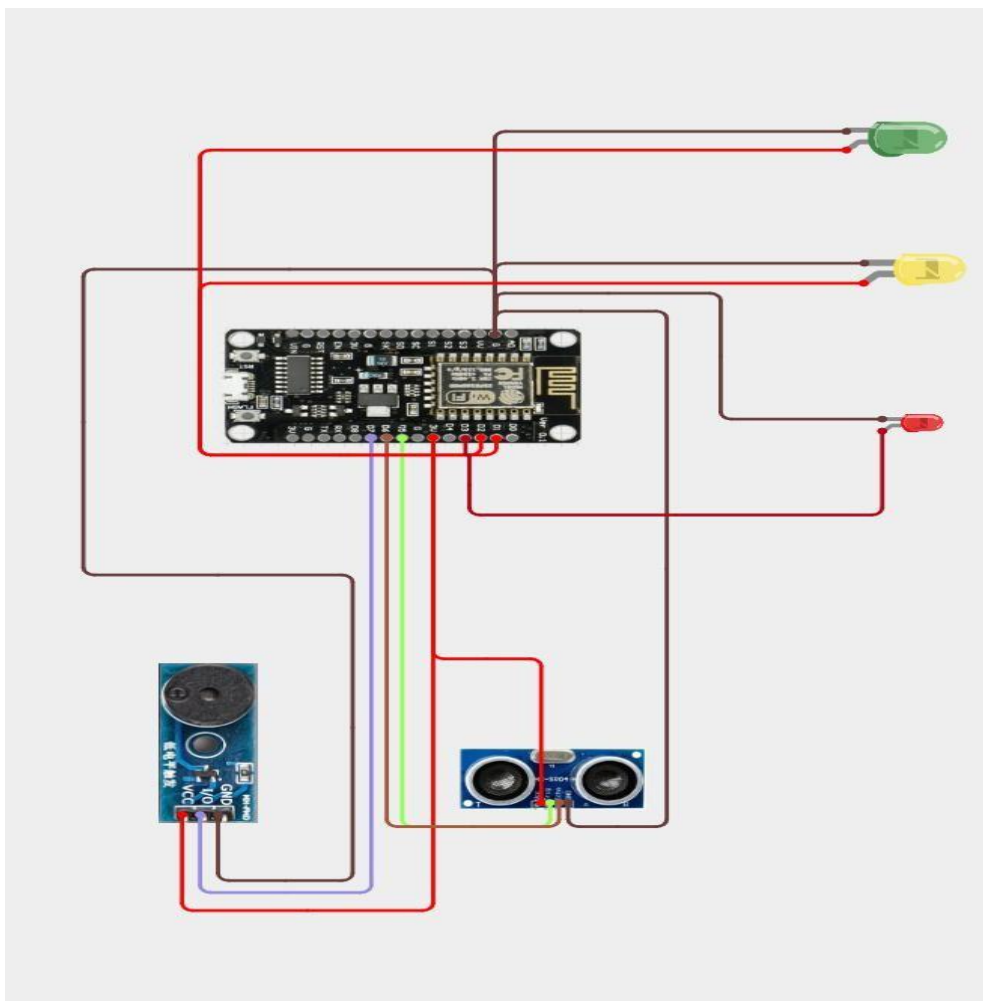
4. SYSTEM ARCHITECTURE – FLOW CHART



CIRCUIT TABLE

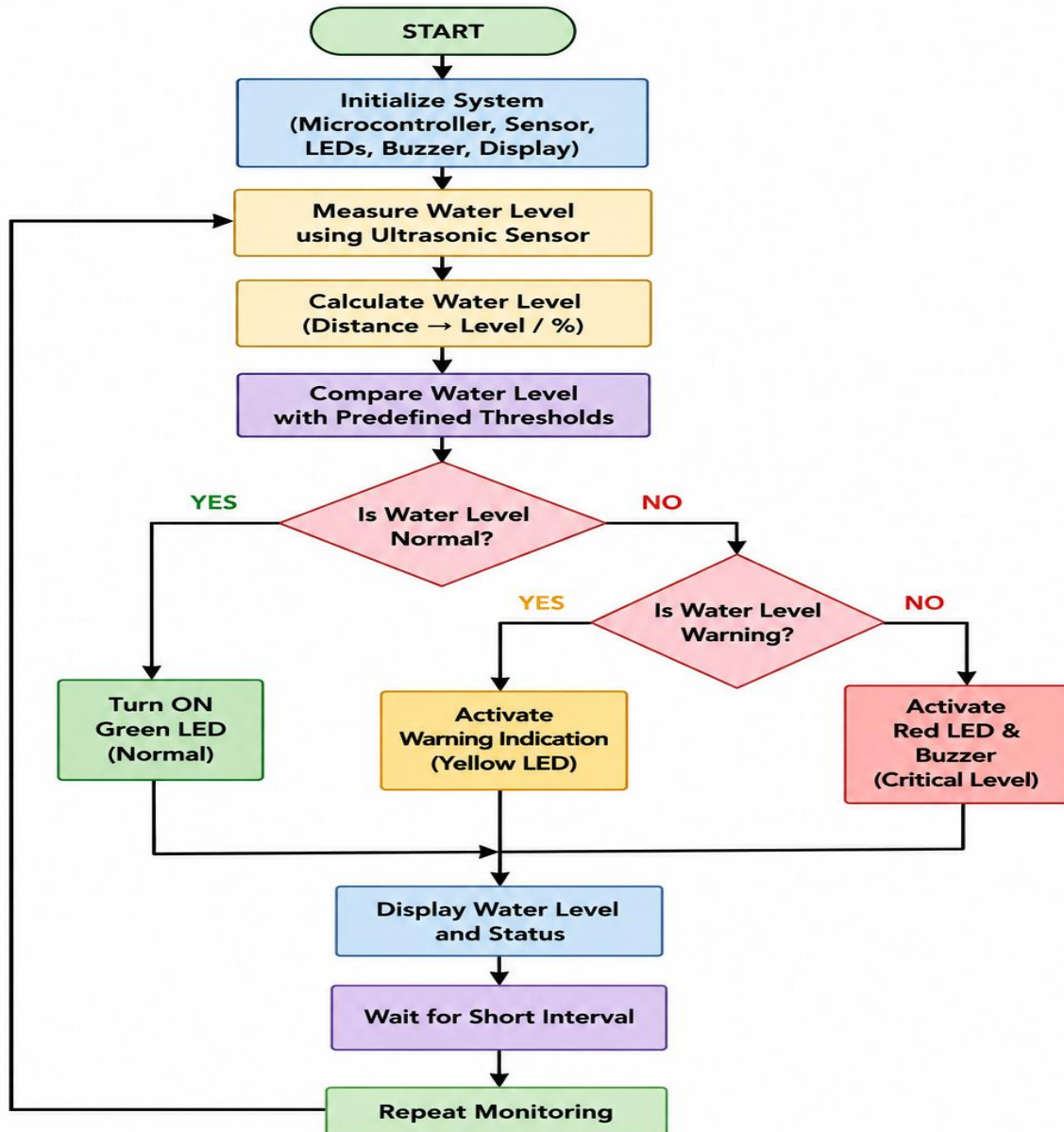
COMPONENT	PIN	NodeMCU
HC-SR04	TRIG	D5
HC-SR04	ECHO	D6
OLED	SDA	D2
OLED	SCL	D1
GREEN LED	SINGAL	D0
RED LED	SINGAL	D4
BUZZER	IN	D7
ALL	GND	GNG

CIRCUIT DESIGN



ALGORITHM

ALGORITHM FLOW CHART



- Start the System
- Initialize the microcontroller, ultrasonic sensor, LEDs, buzzer, and display.
- Measure the distance to the water surface using the ultrasonic sensor.
- Calculate the corresponding water level.
- Compare the water level with the predefined threshold values.
- If the level is normal, turn ON the green LED.
- If the level reaches the warning range, activate the warning indication.
- If the level reaches the critical range, turn ON the red LED and buzzer.
- Display the water level and current status.
- Wait for a short interval.
- Repeat the monitoring process continuously.

CODING

```
H, 30000);
```

```
/******
```

```
Water Level Detection & Flood Alert System NodeMCU
```

```
ESP8266
```

```
*****/
```

```
#include <ESP8266WiFi.h> #include
```

```
"AdafruitIO_WiFi.h"
```

```
// WiFi Credentials
```

```
#define WIFI_SSID "YOUR_WIFI_NAME" #define
```

```
WIFI_PASS "YOUR_WIFI_PASSWORD"
```

```
// Adafruit IO Credentials
```

```
#define IO_USERNAME "YOUR_ADAFRUIT_USERNAME" #define IO_KEY
```

```
"YOUR_ADAFRUIT_KEY"
```

```
AdafruitIO_WiFi io(IO_USERNAME, IO_KEY, WIFI_SSID, WIFI_PASS);
```

```
// Adafruit IO Feeds
```

```
AdafruitIO_Feed *waterLevelFeed = io.feed("water-level"); AdafruitIO_Feed
```

```
*distanceFeed = io.feed("distance"); AdafruitIO_Feed *statusFeed =
```

```
io.feed("status");
```

```
// GPIO Pins
```

```
#define TRIG_PIN          14  // GPIO14 (D5)
```

```
#define ECHO_PIN          12  // GPIO12 (D6)
```

```
#define GREEN_LED         5   // GPIO5 (D1)
```

```
#define YELLOW_LED        4   // GPIO4 (D2)
```

```
#define RED_LED           13  // GPIO13 (D7)
```

```
#define BUZZER            15  // GPIO15 (D8)
```

```
// Tank Height
```

```
const float tankHeight = 100.0;          // cm
```

```
bool alertSent = false; unsigned long
```

```
lastUpload = 0;
```

```
void setup() {
```

```
  Serial.begin(115200);
```

```
  pinMode(TRIG_PIN, OUTPUT); pinMode(ECHO_PIN,  
  INPUT);
```



```
pinMode(GREEN_LED, OUTPUT); pinMode(YELLOW_LED,  
OUTPUT); pinMode(RED_LED, OUTPUT);  
pinMode(BUZZER, OUTPUT);
```

```
digitalWrite(GREEN_LED, LOW);  
digitalWrite(YELLOW_LED, LOW);  
digitalWrite(RED_LED, LOW);  
digitalWrite(BUZZER, LOW);
```

```
Serial.println("Connecting to Adafruit IO...");
```

```
io.connect();
```

```
while (io.status() < AIO_CONNECTED) { Serial.print(".");  
delay(500);  
}
```

```
Serial.println();  
Serial.println("Adafruit IO Connected!");  
}
```

```
void loop() {
```

```
io.run();
```

```
float distance = readDistance();
```

```
float level = ((tankHeight - distance) / tankHeight) * 100.0; level =  
constrain(level, 0, 100);
```

```
String status = "SAFE";
```

```
if (level < 40) {
```

```
digitalWrite(GREEN_LED, HIGH);
```

```
digitalWrite(YELLOW_LED, LOW);
```

```
digitalWrite(RED_LED, LOW);
```

```
digitalWrite(BUZZER, LOW);
```

```
status = "SAFE"; alertSent = false;
```

```
}
```

```
else if (level < 80) {
```

```
digitalWrite(GREEN_LED, LOW);
```

```
digitalWrite(YELLOW_LED, HIGH);
```

```
digitalWrite(RED_LED, LOW);
```

```
digitalWrite(BUZZER, LOW);
```

```
status = "WARNING"; alertSent =
```

```
false;
```

```
}
```

```
else {
```

```
digitalWrite(GREEN_LED, LOW);
```

```
digitalWrite(YELLOW_LED, LOW);
```

```
digitalWrite(RED_LED, HIGH);
```

```
digitalWrite(BUZZER, HIGH);
```

```
status = "DANGER";
```

```
if (!alertSent) {
```

```
Serial.println("SMS Sent: Danger Alert"); alertSent = true;
```

```
}
```

```
}
```

```
// Serial Monitor Serial.print("Distance:
```

```
");
```

```
Serial.print(distance, 1); Serial.print("
cm | Level: "); Serial.print(level, 1);
Serial.print("% | Status: ");
Serial.println(status);
```

```
// Upload to Adafruit IO every 5 seconds if (millis() -
lastUpload >= 5000) {
```

```
waterLevelFeed->save(level);
distanceFeed->save(distance); statusFeed-
>save(status);
```

```
lastUpload = millis();
}
```

```
delay(300);
}
```

```
// Ultrasonic Function float
readDistance() {
```

```
digitalWrite(TRIG_PIN, LOW);
delayMicroseconds(2);
```

```
digitalWrite(TRIG_PIN, HIGH);  
delayMicroseconds(10);  
  
digitalWrite(TRIG_PIN, LOW);  
  
long duration = pulseIn(ECHO_PIN, HIGH);  
  
if (duration == 0) return  
    tankHeight;  
  
return duration * 0.0343 / 2.0;  
} Serial.println(status);  
  
delay(1000);  
}
```

SYSTEM WORKING

The system works by continuously measuring the water level using an ultrasonic sensor placed above the water surface. The sensor sends ultrasonic waves and receives the reflected signal from the water surface. The microcontroller calculates the distance and determines the corresponding water level. The measured level is compared with predefined safety limits. Under normal conditions, the green LED indicates a safe level. When the water level rises to the warning range, a warning indication is activated. When the water reaches the critical level, the red LED and buzzer are activated to provide an immediate flood alert. The water level and status are displayed, and the system continuously repeats the monitoring process for real-time flood detection.

In addition, the system provides real-time information about the water level through the display and monitoring dashboard. The collected readings can be observed continuously to identify sudden increases in water level and take necessary preventive actions. By combining automatic sensing with visual and audible alerts, the system reduces the need for constant manual monitoring and improves the response time during potential flood conditions. The system can also be further enhanced with IoT connectivity to send remote notifications and store water-level data for future analysis.

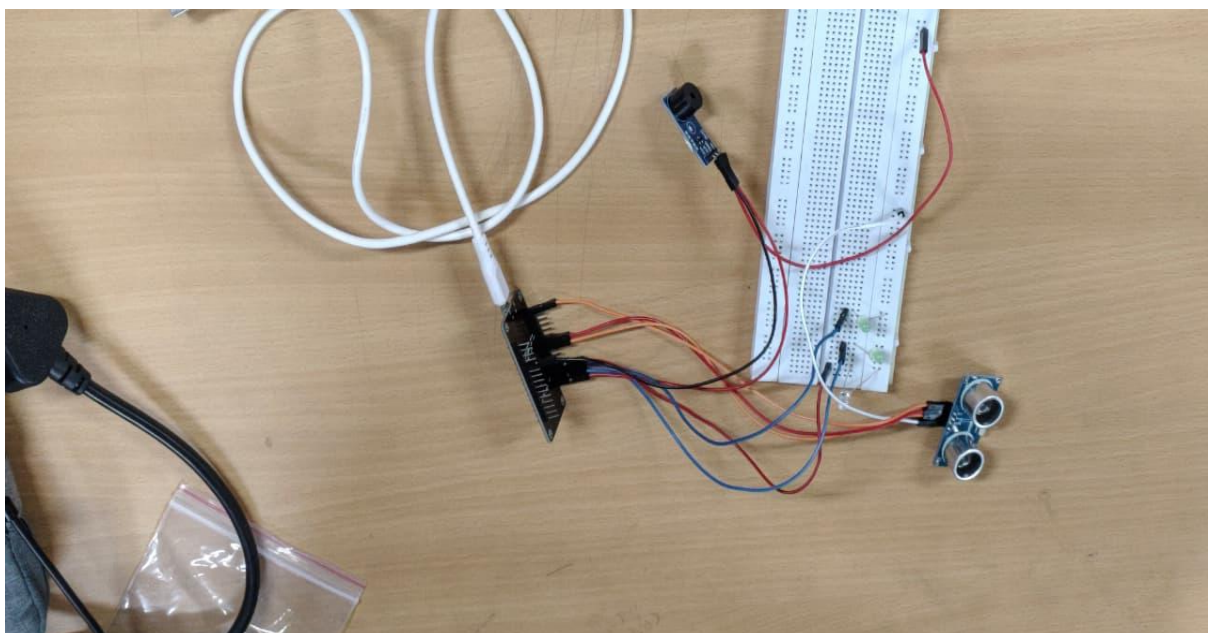
WORKING SEQUENCE

Water Surface → HC-SR04 Ultrasonic Sensor → Echo Signal Reception → Arduino Uno Processing → Water Level Calculation → Threshold Check → Green LED (Normal) / Yellow LED (Warning) / Red LED + Buzzer (Critical) → Continuous Monitoring

BILL OF MATERIALS

S.NO	COMPONENT	QUANTITY
1	ESP-8266 NodeMCU	1
2	HC-SR04 ULTRASONIC SENSOR	1
3	GREEN LED	1
4	YELLOW LED	1
5	RED LED	1
6	BUZZER	1
7	BREADBOARD	1
8	JUMPER WIRES	1 SET
9	5V USB POWER SUPPLY	1

PROTOTYPE IMPLEMENTATION



RESULT AND PERFORMANC ANALYSIS

The Water Level Detection and Flood Alert System was successfully designed, implemented, and tested for real-time water-level monitoring. The ultrasonic sensor continuously measured the distance between the sensor and the water surface, and the microcontroller processed the readings to determine the water level. The system displayed the measured water level and status through the OLED display and web dashboard.

During testing, the system was evaluated under normal, warning, and critical water-level conditions. When the water level remained within the safe range, the green LED indicated normal operation. When the water level increased to the warning range, the warning indication was activated. When the water level reached the critical condition, the red LED and buzzer were activated immediately to provide a flood alert. During the test, the dashboard displayed a water-level reading of approximately **98.98%**, indicating a critical or danger condition.

Overall, the results demonstrate that the proposed system can provide reliable water-level monitoring and timely flood warnings. The combination of ultrasonic sensing, local LED and buzzer alerts, OLED display, and remote dashboard monitoring makes the system suitable for applications such as rivers, tanks, dams, reservoirs, and other flood-prone areas. The system is simple, cost-effective, and can be further improved by adding mobile notifications, cloud data storage, and advanced IoT-based monitoring.

Advantages:

- Non-Contact Measurement
- Low-Cost & Simple Design
- Multi-Level Early Alerting
- Remote Monitoring & Cloud Integration
- System Scalability

Limitations:

- Environmental & Surface Sensitivity
- Network & Internet Dependency
- Lack of Direct Automated Control

Sensor Range & Precision Limits